3. Requirements Gathering Specification

Digital Egypt Pioneers Initiative (DEPI) • Graduation Project Documentation

Project Name: Unstuck

Project: AI Freelance Job Finder & Assistant (AI Automation / Freelance Intelligence Platform)

TRACK & PROGRAM	TARGET MILESTONE	DOCUMENT CODE	ACADEMIC CYCLE
DEPI Technical Track	16 October 2026	DEPI-DOC-03 (M3)	October – December 2026

1. Stakeholder Analysis

Stakeholder Group	Profile & Key Pain Points	Value Delivered by System
Active Freelancers *(Primary Stakeholder)*	Technical freelancers (Software, Data, AI, DevOps) bidding actively across multiple platforms; fatigued by manual feed refreshing; misses deals due to delayed discovery.	Unified real-time opportunity feed, explainable AI compatibility matching, automated proposal drafting, and autonomous bidding.
Freelance Beginners & Upskillers *(Secondary Stakeholder)*	Junior developers building portfolios; uncertain whether their current competencies qualify for client postings; struggles with proposal copywriting.	Clear skill-gap visualization itemizing missing competencies, upskilling recommendations, and tailored cover letter suggestions.
DEPI Evaluation Committee *(Academic Stakeholder)*	Technical supervisors and university professors assessing graduation projects against software engineering rigor, architecture, and innovation.	Strict compliance with DEPI 6-milestone timeline, normalized database schemas, modular clean code, and documented QA verification.

2. User Stories & Agile Use Cases

2.1 User Stories

Story ID	As a...	I want to...	So that...
US-01	Freelancer	Have freelance listings collected automatically from Khamsat, Mostaql, Nafezly, and Freelancer	I eliminate hours of manual switching and refreshing across websites.
US-02	Freelancer	Maintain a structured technical skill profile and portfolio summary	The AI engine accurately evaluates my qualification against incoming opportunities.
US-03	Freelancer	Inspect a 0–100% compatibility score with matched and missing skill badges	I can immediately prioritize high-probability opportunities and address gaps.
US-04	Freelancer	Generate a customized proposal and tailored cover letter with one click	I submit compelling, professional bids quickly without writer's block.
US-05	Freelancer	Enable autonomous smart bidding with custom price thresholds and manual oversight	High-matching jobs are applied to automatically even when I am away from my desk.
US-06	Freelancer	Explore market analytics showing top in-demand skills, average budgets, and posting volume	I can make data-driven career and pricing decisions.
US-07	Freelancer	Filter and search opportunities by budget, platform, keyword, and date	I quickly isolate projects that match my desired compensation and niche.

2.2 Main Operational Use Cases

Use Case 1: Automated Multi-Platform Harvesting & Deduplication (System Process)

Trigger: Scheduled n8n cron workflow or administrative trigger.
Flow: Scrapers invoke Playwright stealth workers across target portals → Extract Title, Description, Budget, Skills, URL → Generate MD5 content hash → Check PostgreSQL database for existing records → Ingest new unique jobs → Write batch audit metrics to `ingestion_logs`.

Use Case 2: AI Semantic Skill-Gap Evaluation & Proposal Copywriting (User-Initiated)

Trigger: Freelancer clicks "Evaluate Match & Draft Proposal" on a job card.
Flow: Frontend sends request to `POST /api/jobs/{id}/evaluate` → Backend retrieves job data and freelancer profile →

Invokes LLM prompt with structured Pydantic schema → Returns match percentage, matched skills, missing skills, and tailored cover letter → Persists result in `match_evaluations` → Displays in UI modal.

Use Case 3: Autonomous Application Submission (Auto-Apply Pipeline)
Trigger: High match threshold exceeded (≥ 85%) with auto-apply enabled, or user clicks "Auto Apply".
Flow: Multi-Agent coordinator calculates optimal bid price based on budget & profile → Assembles proposal text → Submits bid via platform API or automated browser session → Logs submission receipt and updates job status to `APPLIED`.

3. Functional Requirements (FRs)

Module	Req ID	Functional Requirement Specification	Priority
Data Collection & Ingestion	**FR-1.1**	System shall harvest job listings across Khamsat, Mostaql, Nafezly, Freelancer, and Upwork.	**Must-Have**
	FR-1.2	System shall bypass anti-bot and rate-limiting using proxy rotation and session persistence.	**Must-Have**
	FR-1.3	System shall enforce 100% hash deduplication and currency normalization before ingestion.	**Must-Have**
Database & Storage	**FR-2.1**	PostgreSQL database shall store Jobs, Profiles, Skills, Matches, Proposals, and Logs.	**Must-Have**
	FR-2.2	System shall cache frequent query results and AI matches in Redis for sub-second retrieval.	**Should-Have**
Web Platform & APIs	**FR-3.1**	FastAPI backend shall expose RESTful endpoints for listing, filtering, pagination, and analytics.	**Must-Have**
	FR-3.2	React frontend shall offer responsive RTL/LTR interfaces with keyword search and dynamic filters.	**Must-Have**
	FR-3.3	Frontend shall include a Market Analytics Dashboard visualizing volume, budgets, and skill trends.	**Must-Have**
AI Matching & Proposal Engine	**FR-4.1**	AI engine shall compute semantic skill overlap and output a 0–100% explainable compatibility score.	**Must-Have**
	FR-4.2	AI copywriter shall generate tailored proposal text and custom cover letters personalized to the job.	**Must-Have**
	FR-4.3	Autonomous submission engine shall calculate optimal bid pricing and execute application workflows.	**Must-Have**

4. Non-Functional Requirements (NFRs)

Category	NFR ID	Specification & Measurable Target Metric
Performance	**NFR-01**	**API Latency:** Filter and search database endpoints must return responses in < 250 milliseconds.
Throughput	**NFR-02**	**Batch Ingestion:** Pipeline must harvest, sanitize, deduplicate, and persist 50 jobs in < 60 seconds.
Security & Privacy	**NFR-03**	**Credential Protection:** All AI keys, proxy credentials, and platform secrets must reside strictly in environment variables; zero plaintext commits.
Reliability	**NFR-04**	**Fault Tolerance:** Platform scraper network timeouts or DOM errors must fail gracefully without corrupting database integrity.
AI Schema Validity	**NFR-05**	**Strict Structured Output:** 100% of LLM evaluations must conform strictly to Pydantic JSON schemas to guarantee zero frontend runtime errors.
Maintainability	**NFR-06**	**Clean Separation of Concerns:** Independent micro-layering between Ingestion, Database, API Server, AI Agent, and Web Frontend.

5. Traceability Matrix to Official DEPI Milestones

DEPI Official Milestone	Official Target Date	Associated Requirements & Technical Artifacts
Project Planning & Management	16 October 2026	System architecture, team RACI matrix, risk assessment, and 100% in-scope feature boundaries.

DEPI Official Milestone	Official Target Date	Associated Requirements & Technical Artifacts
Literature Review	16 October 2026	Market landscape analysis, competitor benchmarking, and technological stack evaluation.
Requirements Gathering	16 October 2026	User stories (US-01 to US-07), functional requirements (FR-1 to FR-4), and measurable NFRs.
System Analysis & Design	06 November 2026	UML diagrams, database ERD schema, sequence diagrams, and Multi-Agent workflow definitions.
Implementation (Source Code)	30 November 2026	Full working source code: n8n pipelines, Playwright scrapers, FastAPI backend, React dashboard, AI agents.
Final Presentation & Reports	04 December 2026	Comprehensive final documentation, QA test reports, slide deck presentation, and live project defense.